

EDUCATION

University of California, Los Angeles

Ph.D in Mathematics with committee chair: Mike Hill

Los Angeles, California

2023

University of Michigan

B.S. Mathematics with Honors

Ann Arbor, MI

2017

EXPERIENCE

Postdoctoral Assistant Professor of Mathematics

University of Michigan

August 2023 - July 2026

PUBLICATIONS

Paul Reschke and Bar Roytman. “Lower semicontinuity of entropy in a family of K3 surface automorphisms.” *Rocky Mountain Journal of Mathematics* 47 (2017), no. 7, 2323–2349. doi:10.1216/RMJ-2017-47-7-2323

PREPRINTS

Po Hu, Igor Kriz, Bar Roytman, and Petr Somberg. “Périodicités globales d’homologie équivariante ordinaire.” **Submitted**, 2025.

Igor Kriz and Bar Roytman. “What is an equivariant Adams spectral sequence?” **Preprint**, 2025. PDF:

<https://krizigor.github.io/pushforward202506.pdf>

MANUSCRIPTS IN PREPARATION

Jiajun Cheng, Beinu Guo, Haran Mouli, and Bar Roytman. “The realizability problem for linear isometries and Steiner operads over cyclic groups.” **In preparation**, 2026.

Bar Roytman. “Operads and their algebras in equivariant stable homotopy theories.” **In preparation**, 2026.

Bar Roytman. “Highly structured orientations from equivariant Thom spectra.” **In preparation**, 2026.

DISSERTATION

Bar Roytman. “Highly structured orientations from equivariant Thom spectra.” Ph.D. thesis, University of California, Los Angeles, **2023**. Advisor: Michael A. Hill. <https://escholarship.org/uc/item/8jc7s298>

INVITED TALKS

Parameterized proper equivariant incomplete

Joint Mathematics Meetings January 11, 2025

stable homotopy theories and their operadic algebras

Geometry of Operads in Equivariant Homotopical Algebra

Vanderbilt University April 19, 2024

Highly Structured Real Orientations

University of Chicago October 31, 2023

Obstructions to maps out of equivariant multiplicative Thom spectra

FRG # 2052042 Dec 5, 2022

The Materials of Mathematics

University of Michigan Math Club Apr 22, 2021

TEACHING EXPERIENCE (AS UNIVERSITY OF MICHIGAN INSTRUCTOR OF RECORD)

Math 217: Linear Algebra

Math 214: Linear Algebra with Applications

Math 116: Calculus II

TEACHING (AS UCLA TEACHING ASSISTANT)

Math 225B: Differential Geometry

Math 61: Discrete Structures

Math 115AH: Linear Algebra (Honors)

Math 225C: Algebraic Topology

Math 121: Introduction to Topology

Math 115A: Linear Algebra

Math 111: Theory of Numbers

Math 123: Foundations of Geometry

Math 174E: Mathematics of Finance for Mathematics/Economics Students

Math 110A: Algebra

Math 31B: Integration and Infinite Series

Math 32B: Calculus of Several Variables

Math 31A: Differential and Integral Calculus

Math 170A: Probability Theory

SERVICE

REU Mentorship

University of Michigan

Mentored eleven students in REUs about transfer systems associated to linear isometry and Steiner operads, and weak transfer systems.

Summers 2024, 2025, 2026

Lab of Geometry at Michigan (LoG(M)) Mentorship

University of Michigan

Faculty mentor for four students taking Math 440, a course on mathematical research skills. Students wrote expository, research documents, and slides in LaTeX, gave talks, and worked on a project investigating transfer systems associated to linear isometry operads.

Fall 2024

Instructor at U(M) Math Corps

Washtenaw County, Michigan

Ran courses on mathematical proof, combinatorial game theory, and algebra for the high school volunteers of the U(M) Math Corps program. Developed and modified instructional materials for the latter two courses. The courses emphasizes high standards, was well-staffed to provide real-time and remote feedback, and were designed to be taken multiple times, enjoyably.

Summers 2021, 2023, 2024

Directed Reading Program

UCLA Mathematics

Volunteer individualized reading course mentorship program. My reading courses on number theory, manifold theory, algebraic topology, and homological algebra.

2019-2022

AWARDS

Horn-Moez Prize

University of California, Los Angeles

Excellence in first year graduate studies.

June 2017

Outstanding Graduating Senior Award

University of Michigan, Ann Arbor

May 2016

Frank Raymond Award in Geometry and Topology

University of Michigan, Ann Arbor

May 2013